Karunakar Cheekolu

AI Engineer | Generative AI, LLM Agents & RAG | AWS

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SUMMARY

AI Engineer with 2+ years of experience building production Generative AI (GenAI) systems on AWS: Amazon Bedrock (Claude) large language model (LLM) agents, multi-agent systems, hybrid Retrieval-Augmented Generation (RAG), LLM evaluation and guardrails. Delivered 15+ AI and full-stack applications across manufacturing, procurement, healthcare, construction and marketing, including a public RAG assistant over Indian building codes and an email-driven procurement agent. M.Tech in AI & Data Science from Christ University (CGPA 9.1/10), B.Tech in Computer Science Engineering from IIT Dharwad, and Springer-published researcher.

TECHNICAL SKILLS

Generative AI: LLMs, Amazon Bedrock (Claude), GPT, Gemini, Llama, Prompt Engineering, Structured Outputs, Guardrails
Agents & RAG: AI Agents, Multi-Agent Systems, Human-in-the-Loop, RAG, Hybrid Search, FAISS, Qdrant, LLM Evaluation
Machine Learning: PyTorch, Hugging Face Transformers, scikit-learn, NLP, Computer Vision, OpenCV
Languages: Python, TypeScript, JavaScript, SQL
Backend & Frontend: FastAPI, Flask, Node.js, REST APIs, Next.js, React
Cloud & DevOps: AWS (EC2, Bedrock, Lambda, API Gateway, S3, DynamoDB, Cognito, IAM), Docker, Nginx, Linux, CI/CD
Data & Tools: PostgreSQL, DynamoDB, Neo4j, SQLite, Pandas, Power BI, Git, GitHub Actions, Claude Code, Codex

EXPERIENCE

AI Engineer (SDE-1) **Aug 2025 – Present**
Aispresso Labs

- Own end-to-end delivery of 15+ AI and full-stack applications (Python, FastAPI, React, Next.js), 10 in production, from discovery and architecture to AWS deployment, runbooks and client handoff.
- Integrated Amazon Bedrock (Claude) into 5+ workflows (invoice extraction, catalogue vision extraction, brief parsing, claim assessment), with draft/final model tiers and a Bedrock/Gemini/Groq provider adapter.
- Established LLM guardrails across projects: page-cited RAG answers, numeric-grounding checks, validated LLM extractions, agent evaluation suites and human approval gates on financial and claim decisions.
- Architected and deployed a manufacturing AI Agents Hub of 7 specialist agents (energy, quality, maintenance, warehouse, etc.) as 8 HTTPS services on EC2, with run/event contracts and role-gated approvals.
- Built a 3.7M-node, 6M-edge Neo4j knowledge graph from about 1 GB of Emerson DeltaV control-system exports as the deterministic source of truth for an agentic change-impact platform.
- Planned an Azure-to-AWS migration for an AI SaaS client (13 VMs, 8 container apps, 3 TB MongoDB), including moving its Claude workloads and LiteLLM gateway to Amazon Bedrock.
- Standardized AWS production delivery: EC2 with Docker Compose, Nginx/Caddy TLS, systemd workers, GitHub Actions CI/CD, IAM roles with SSM secrets (no static keys), encrypted S3 and restore-tested backups.

AI & Full-Stack Engineer **Sep 2024 – Apr 2025**
Superchat Inc.

- Integrated and tuned LLM response workflows for AI chatbot products to improve contextual reply quality.
- Built the React/Redux chat app end to end: Google OAuth and JWT sign-in, Firebase integration, LLM API wiring and markdown/code rendering of model replies.

Artificial Intelligence Engineer Intern **Jul 2024 – Jan 2025**
Orsu E Corp

- Built a multilingual translation app on a Transformer sequence-to-sequence model over parallel text corpora, evaluated with BLEU and served through a real-time web interface.

Data Analyst Intern **May 2024 – Jun 2024**
Mentorness

- Analyzed hotel-booking trends in SQL and built a Power BI dashboard of global cost-of-living insights.

PROJECTS

Rivvet: AI Structural Design Assistant | Amazon Bedrock, RAG, Qdrant, FastAPI, React, OR-Tools, AWS

- Built and launched a public AI assistant for Indian residential structural design: Bedrock-powered hybrid RAG (Qdrant + BM25) over IS building codes, returning page-cited answers with source page images.
- Developed a deterministic OR-Tools CP-SAT layout engine that turns a plot and room list into floor plans, column grids and member sizes, with DXF and IFC4 (BIM) export and JWT-secured accounts.

Agentic Procurement Platform | Amazon Bedrock (Claude), FastAPI, React, SQLite, AWS EC2
- Built an email-driven procurement agent that classifies purchasing threads, extracts and validates estimates and invoices with Claude, matches vendors to purchase orders and generates PO documents.
- Deployed web, worker and outbound services as separate systemd units on EC2, with outbound send gates and human sign-off before any PO is sent, invoice approved or payment made.

Multi-Agent Ad Creative Engine | Python, Claude, Gemini Image, SSE, Nginx, AWS EC2
- Built a production multi-agent ad creative engine: a context agent breaks down campaign briefs and a deep agent spawns design and content sub-agents that compose a master prompt for image generation.
- Streamed agent tokens, events and images over SSE to a chat console with a live pipeline trace; deployed on AWS EC2 with systemd, Nginx and a hash-pinned runtime manifest.

Forward-Pricing Supply Intelligence | Node.js, SQLite, Amazon Bedrock, AWS EC2
- Built a pricing platform for a nutrition manufacturer that blends SAP contract and market prices by demand plan into quarterly forward prices for 50+ raw materials.
- Designed a grounded Bedrock assistant: a deterministic rules engine makes each pricing decision, the LLM only explains it, and answers with unsupported numbers are rejected; backed by 550+ tests.

Warranty Genie: AI Claims Platform | Next.js, TypeScript, PostgreSQL, Amazon Bedrock, Docker, AWS
- Built an automotive warranty-claims platform where customers upload invoices, photos and video, Amazon Bedrock pre-assesses the evidence and a human reviewer makes every claim decision.
- Engineered a fault-tolerant background AI worker using PostgreSQL job leases (SKIP LOCKED), lease-token fencing and bounded retries, deployed on EC2 behind Caddy HTTPS.

Multi-Tenant RAG PDF Assistant | Python, FastAPI, FAISS, BM25, Llama, Gemini, AWS EC2, S3
- Built a production multi-tenant RAG platform on AWS for natural-language Q&A over text and visual PDFs, with isolated user knowledge bases, S3-backed assets and persistent document indexes.
- Engineered hybrid FAISS + BM25 retrieval with content hashing, query-aware top-k, page validation, anti-hallucination guardrails and Llama/Gemini fallback.

Additional Projects | Python, Streamlit, scikit-learn, AWS Lambda, DynamoDB, Claude, PyTorch
- Venue recommender lifting recall@5 from 18% to 92% on 82 past proposals; Isolation Forest predictive-maintenance model flagging bearing failure 3 hours before ISO 10816 alarms (NASA IMS data).
- Serverless travel-analytics dashboard (Lambda, DynamoDB, Cognito) with Claude natural-language queries; healthcare patient-lifecycle platform with RBAC and AI search; fine-tuned T5, BERT and YOLOv8.

PUBLICATIONS

Lung Cancer Classification from CT-Scan Images Using Enhanced VGG16 Model | Springer (Scopus Indexed), CML 2025
- Designed a 4-class CT-scan classifier using VGG16 with squeeze-and-excitation blocks and residual connections, achieving over 95% accuracy. DOI: 10.1007/978-981-95-2878-3_33

EDUCATION

Christ University **2025**
Master of Technology in Computer Science Engineering (AI & Data Science) – CGPA: 9.1/10 *Bangalore, India*

Indian Institute of Technology (IIT) Dharwad **2023**
Bachelor of Technology in Computer Science Engineering – CGPA: 6.0/10 *Dharwad, India*

CERTIFICATIONS

Google Cloud Certified Professional Cloud Architect | Google Cloud
AWS Certified Cloud Practitioner | Amazon Web Services